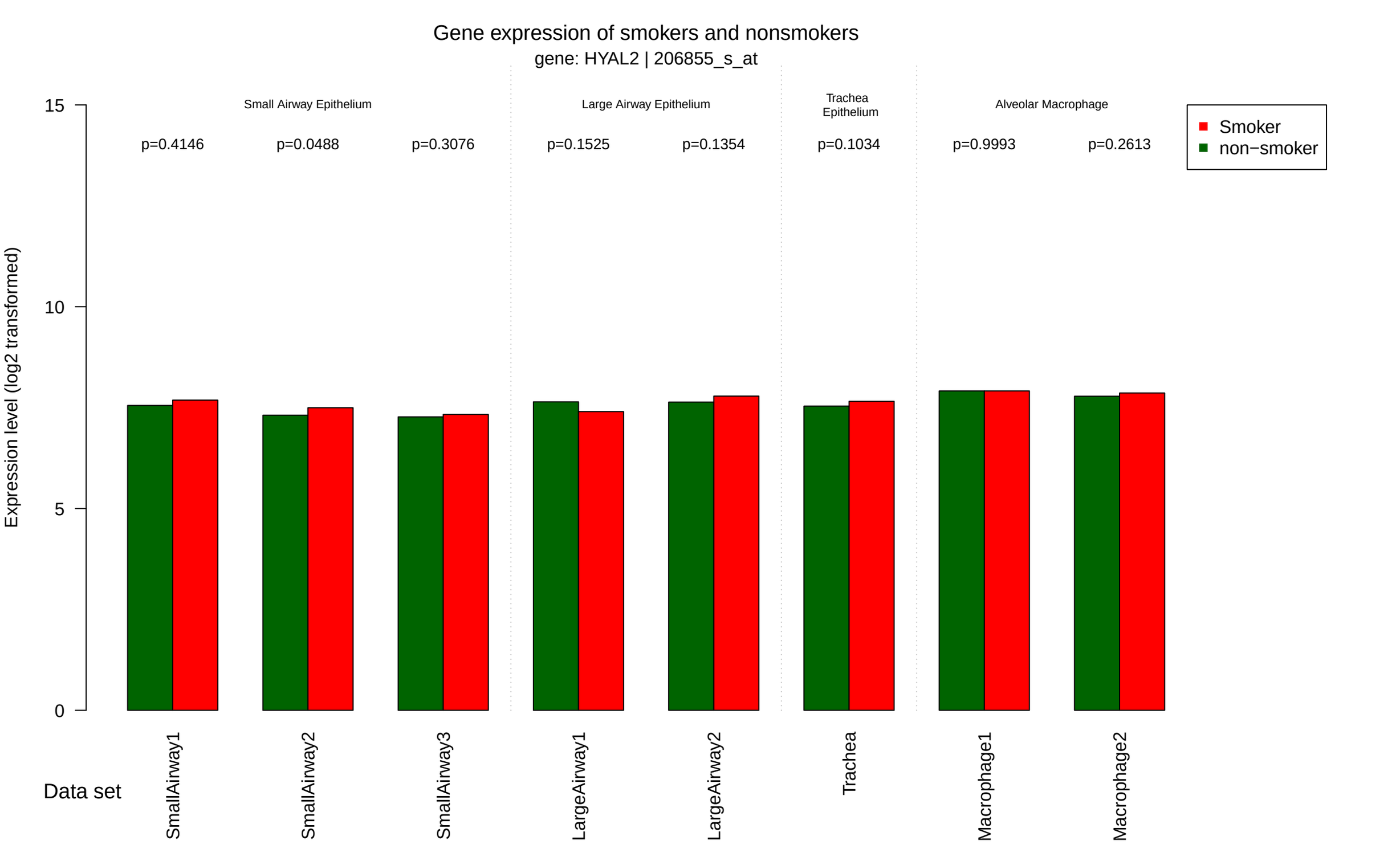
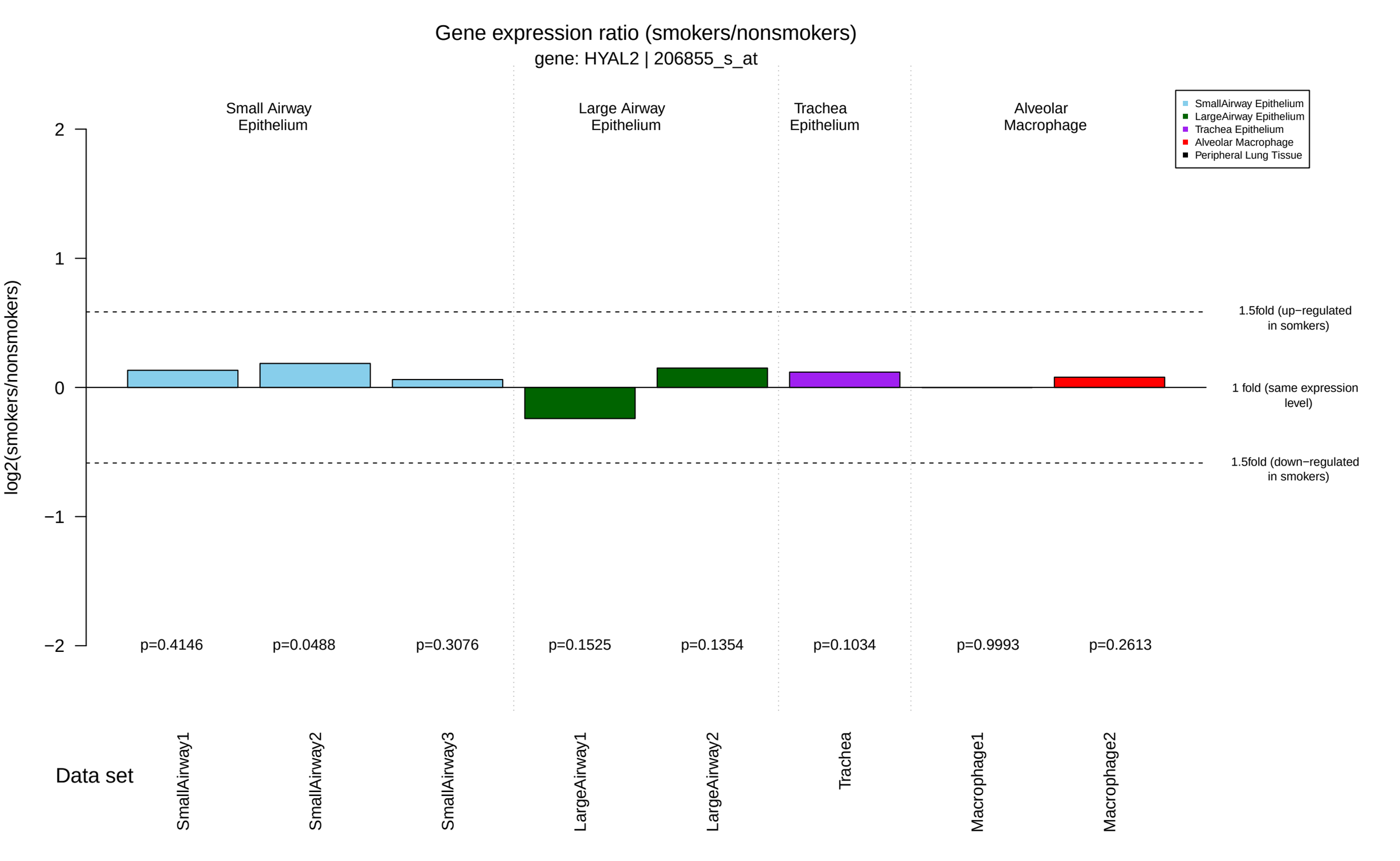
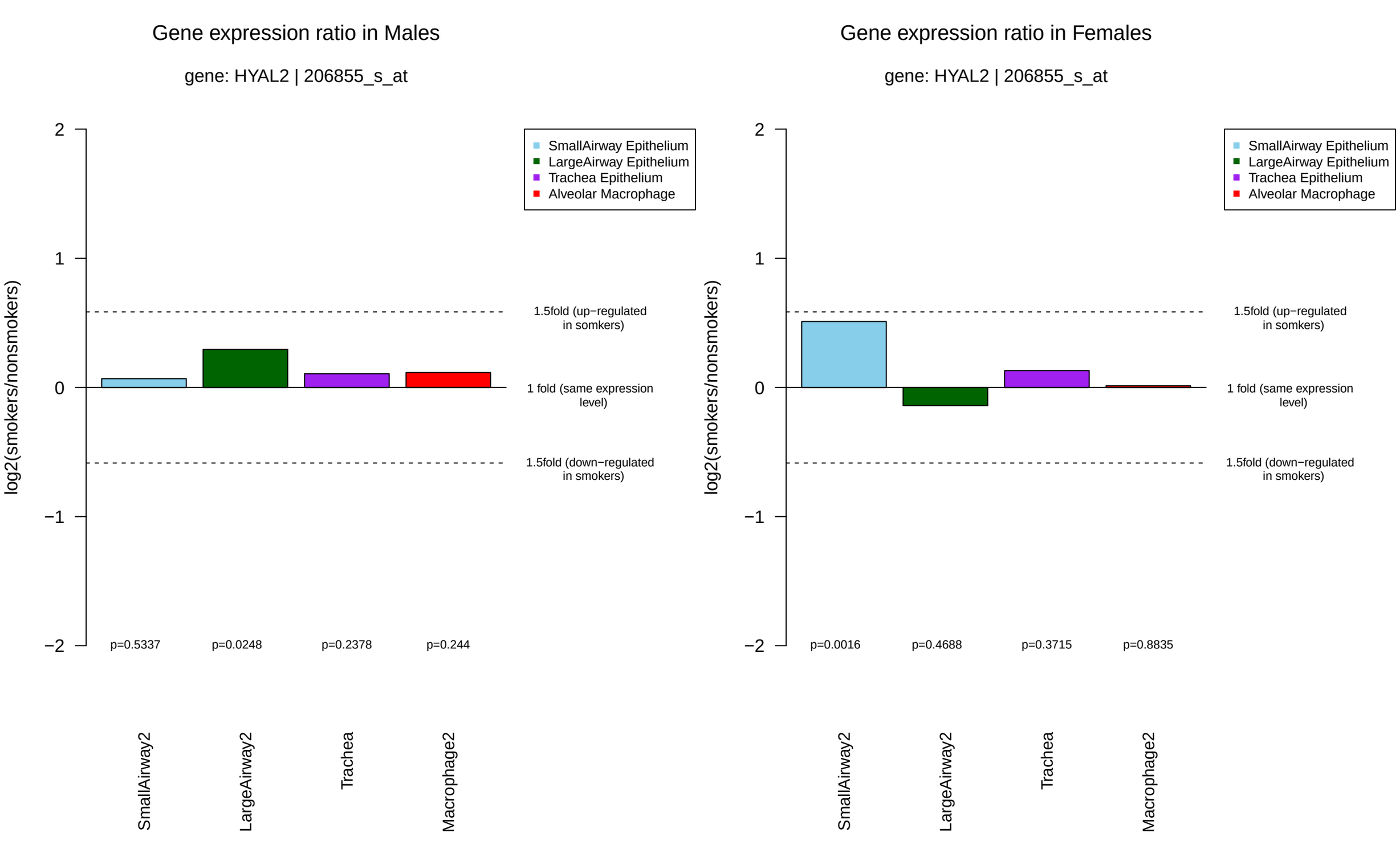
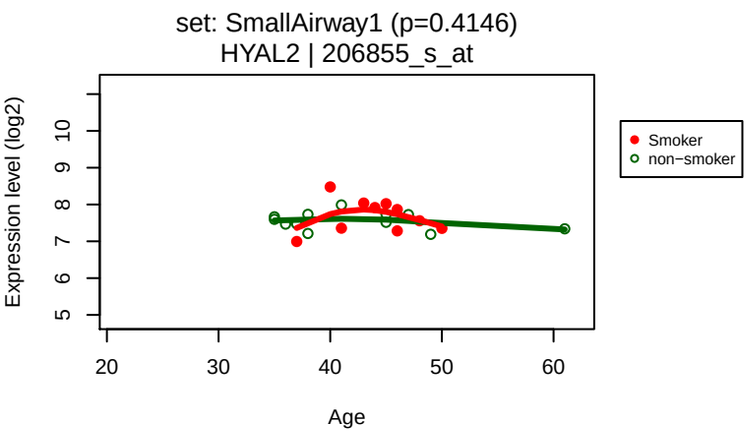
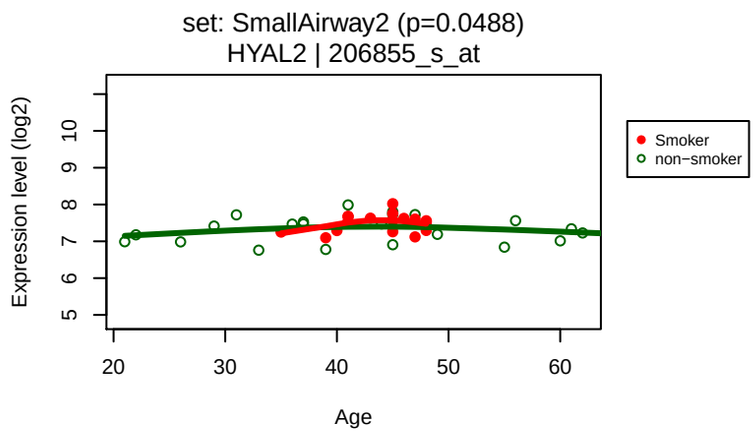
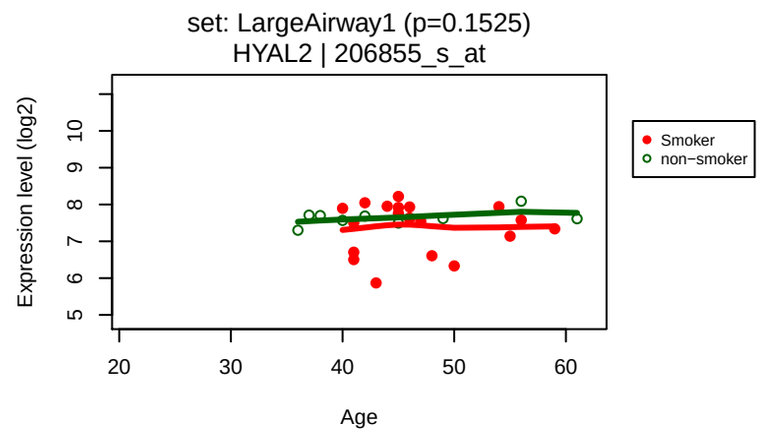

Gene expression of smokers and nonsmokers
gene: HYAL2 | 206855_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4146
p=0.0488
p=0.3076
p=0.1525
p=0.1354
p=0.1034
p=0.9993
p=0.2613
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

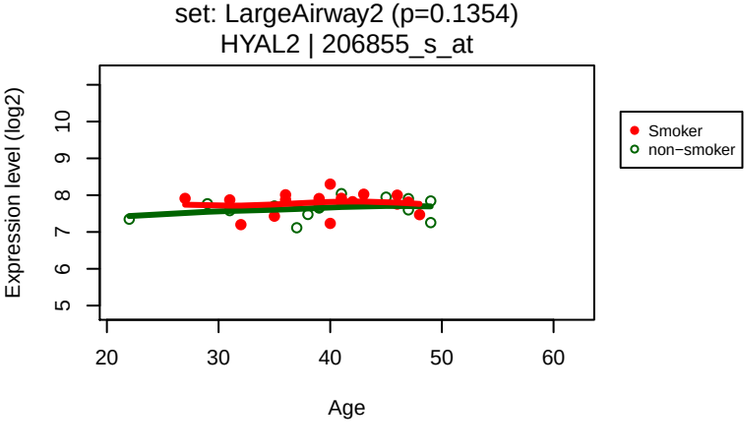

Gene expression ratio (smokers/nonsmokers)
gene: HYAL2 | 206855_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4146
p=0.0488
p=0.3076
p=0.1525
p=0.1354
p=0.1034
p=0.9993
p=0.2613
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

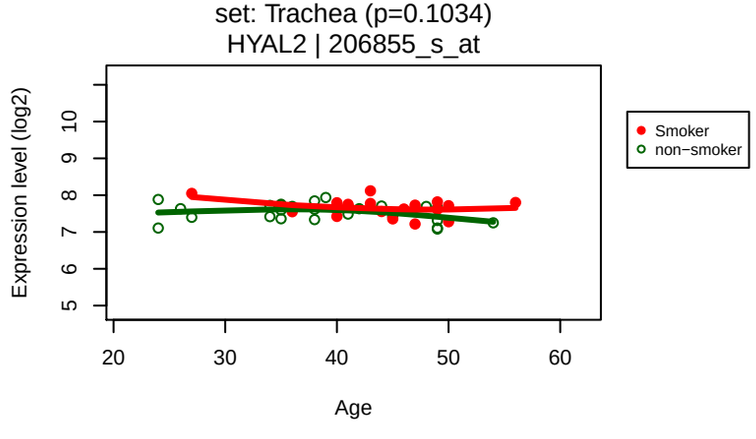

Gene expression ratio in Males
gene: HYAL2 | 206855_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5337
p=0.0248
p=0.2378
p=0.244
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HYAL2 | 206855_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0016
p=0.4688
p=0.3715
p=0.8835
SmallAirway2
LargeAirway2
Trachea
Macrophage2

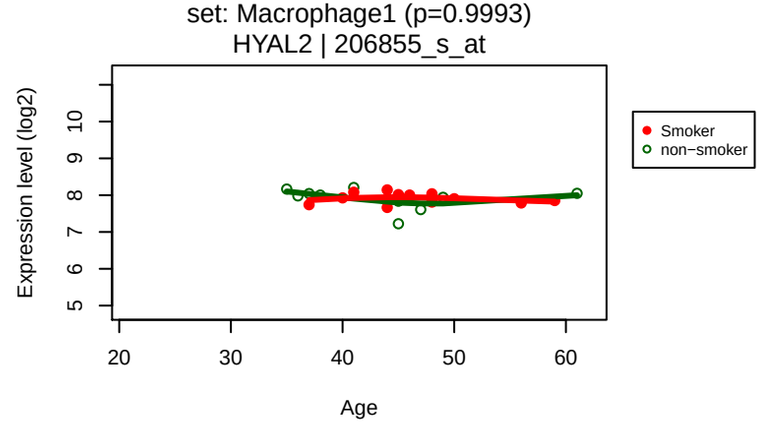

set: SmallAirway1 (p=0.4146)
HYAL2 | 206855_s_at
Expression level (log2)
Age
Smoker
non−smoker

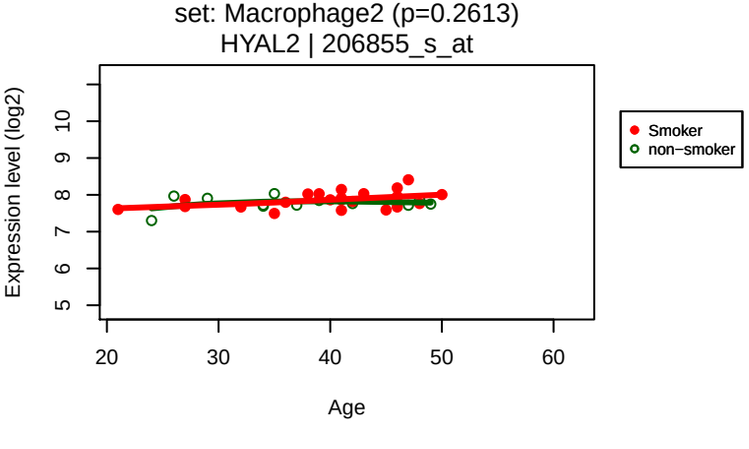

set: SmallAirway2 (p=0.0488)
HYAL2 | 206855_s_at
Expression level (log2)
Age
Smoker
non−smoker

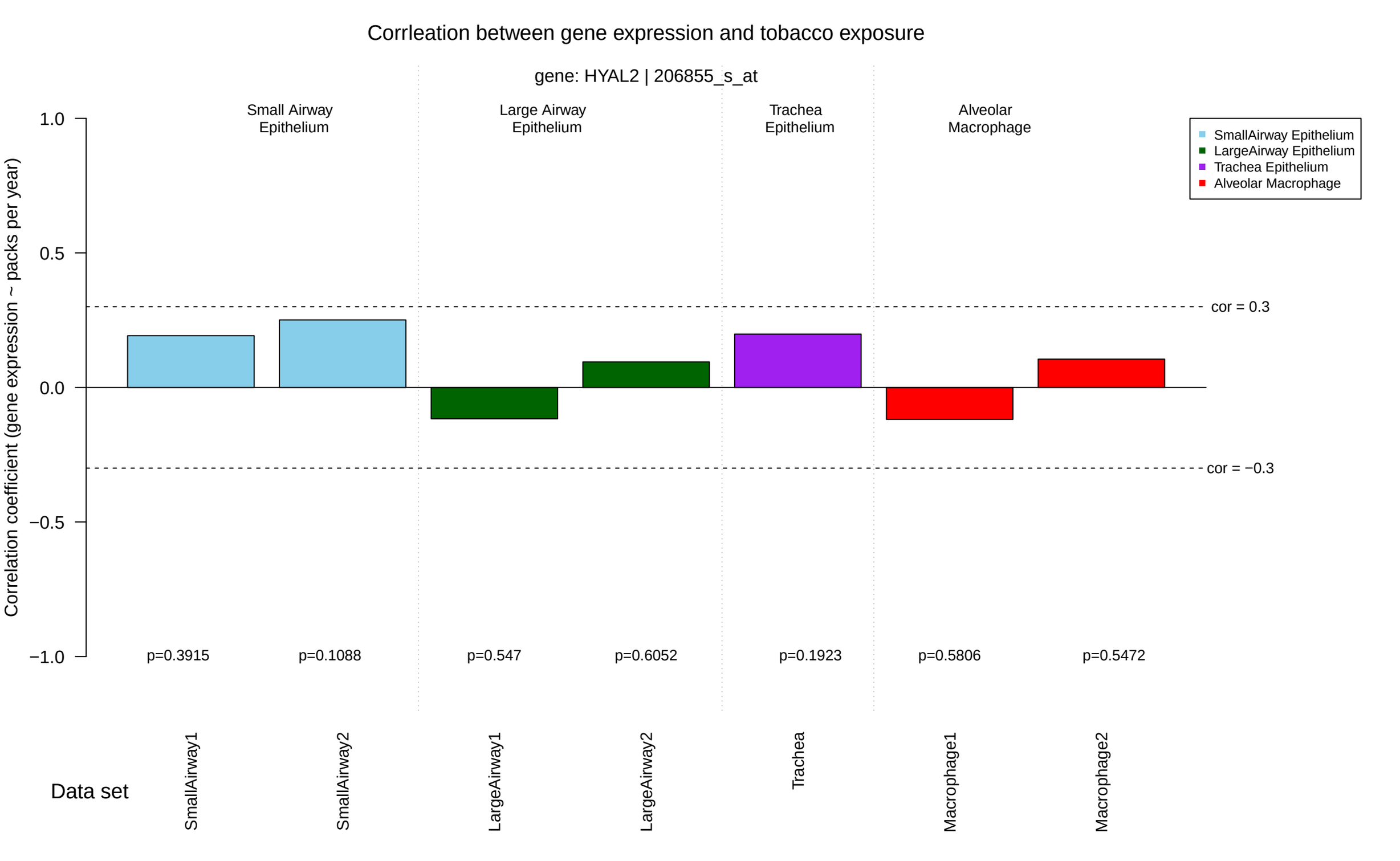

set: LargeAirway1 (p=0.1525)
HYAL2 | 206855_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1354)
HYAL2 | 206855_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1034)
HYAL2 | 206855_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9993)
HYAL2 | 206855_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2613)
HYAL2 | 206855_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HYAL2 | 206855_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3915
p=0.1088
p=0.547
p=0.6052
p=0.1923
p=0.5806
p=0.5472
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2